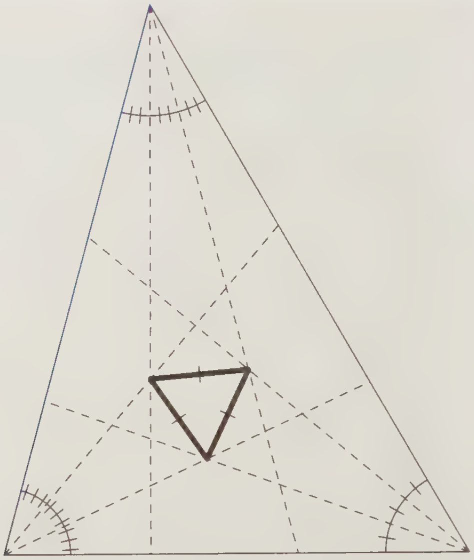


Morley's First Triangle

The worst form of inequality is to try to make unequal things equal. – Aristotle

CHAPTER 10

Geometric Inequalities

In this chapter we will explore three geometric inequalities involving triangles. First, we will examine how the order of the lengths of the sides is related to the order of the measures of the angles in a triangle. Then, we will learn how to tell if a triangle is acute or obtuse just by considering its side lengths. Finally, we will explore the most widely used geometric inequality of all – the Triangle Inequality. Like many of the most useful mathematical tools, the Triangle Inequality is so simple it's almost obvious, but it can be used to develop many non-obvious solutions to complex problems.

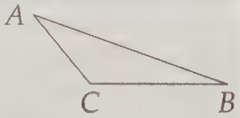
10.1 Sides and Angles of a Triangle

Problems

Problem 10.1: Draw a few triangles. Label each triangle $\triangle ABC$, and compare the order of the lengths AB, BC, and AC (from shortest to longest) to the order of the measures of the angles $\angle A$, $\angle B$, and $\angle C$. Make a guess about how the order of the lengths of the sides is related to the order of the measures of the angles in a triangle. Try to prove your guess always works before continuing this section! (Don't forget, you have to consider acute, right, and obtuse triangles.)

Problem 10.2: Prove that the hypotenuse of a right triangle is longer than each of the other two sides of the triangle.

Problem 10.3: In this problem, we will prove that in an obtuse triangle, the side opposite the obtuse angle is the longest side of the triangle. We start with $\triangle ABC$ with obtuse $\angle C$ as shown.



(a) We already have information about the longest side in a right triangle. Try building a right triangle with $\overline{AB}$ as a hypotenuse to show that $AB > BC$. Can you use a similar approach to show $AB > AC$?

(b) Try finding a second solution by drawing a right triangle with $\overline{BC}$ as a leg.

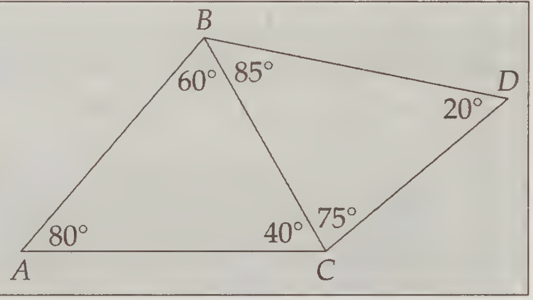
Problem 10.4: Let $\triangle ABC$ be acute with $\angle B > \angle C$. We will show that $AC > AB$.

(a) Draw altitude $\overline{AX}$ and find the point D on $\overleftrightarrow{BC}$ such that $AD = AC$ (C and D are different points). Show that $\angle DAX > \angle BAX$. Use this inequality to show that D is beyond B on $\overrightarrow{XB}$.

(b) Prove that $AC > AB$.

Problem 10.5: Let $\triangle ABC$ be acute. Show that if $AC > AB$, then $\angle B > \angle C$.

Problem 10.6: Given the angles shown in the diagram, order the lengths AB, BD, CD, BC, and AC from least to greatest. Note that the figure is not drawn to scale!



Problem 10.7: In $\triangle ABC$, the median $\overline{AM}$ is longer than $BC/2$. Prove that $\angle BAC$ is acute.

After looking at hundreds, if not thousands, of triangles while working to this point in the book, you probably think that the largest side of a triangle is opposite the triangle's largest angle, and the shortest side is opposite the smallest angle. Good instincts! We'll have to work through a few cases to prove it, though.

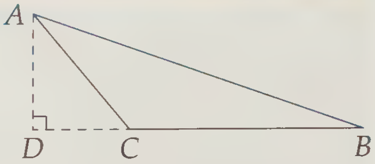
Problem 10.2: Prove that the hypotenuse of a right triangle is longer than each of the other two sides of the triangle.

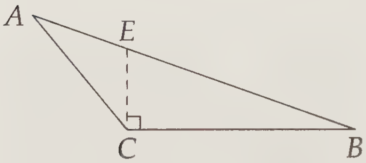
Solution for Problem 10.2: Let $\overline{AC}$ be the hypotenuse of $\triangle ABC$. The Pythagorean Theorem tells us $AC^2 = AB^2 + BC^2$. Since AB^2 and BC^2 are both positive, we must have $AC > AB$ and $AC > BC$. Hence, the hypotenuse of a right triangle is longer than each of the other two sides of the triangle. □

Taking care of the longest side in a right triangle was pretty easy. Now let's take a look at obtuse triangles.

Problem 10.3: Prove that in an obtuse triangle, the side opposite the obtuse angle is the longest side of the triangle.

Solution for Problem 10.3: Let $\angle C$ in $\triangle ABC$ be obtuse. We already know that the hypotenuse of a right triangle is the longest side of the triangle, so we try building a right triangle we can use. We want to show that $\overline{AB}$ is the longest side of our triangle, so we make it the hypotenuse of a right triangle by drawing altitude $\overline{AD}$ from A as shown. Since $\overline{AB}$ is the hypotenuse of $\triangle ABD$, we have $AB > BD$. Since $BD > BC$, we have $AB > BD > BC$. Similarly, we can draw an altitude from B to show that $AB > AC$.





We could also have built a right triangle with $\overline{BC}$ as a leg by drawing a line through C perpendicular to $\overline{BC}$ as shown. Since $\triangle ECB$ is a right triangle, we have $EB > BC$. Since $AB > EB$, we have $AB > EB > BC$. Similarly, we can show that $AB > AC$.

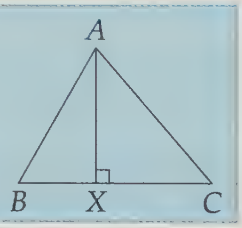
Therefore, in an obtuse triangle, the side opposite the obtuse angle is the longest side of the triangle. □

Now we're ready to deal with acute angles.

Problem 10.4: Let $\triangle ABC$ be acute with $\angle B > \angle C$. Show that $AC > AB$.

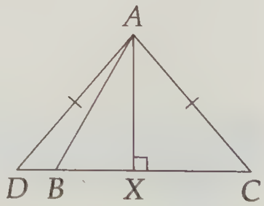
Solution for Problem 10.4: See if you can find what's missing in this 'proof':

Bogus Solution: We start by drawing altitude $\overline{AX}$. Since $\angle B > \angle C$, we have $\angle BAX < \angle CAX$. Therefore, $BX < CX$. Since $AB^2 = AX^2 + BX^2$ and $AC^2 = AX^2 + CX^2$, our $BX < CX$ tells us that $AB < AC$.



The problem here is that we haven't justified that $BX < CX$ claim. Why does $\angle BAX < \angle CAX$ mean that $BX < CX$ (and no, you can't just say 'It's obvious!')?

It's not clear how we can build a right triangle to directly compare AB and AC, but we can build an obtuse one. We draw altitude $\overline{AX}$ and locate point D on $\overleftrightarrow{BC}$ such that $AD = AC$. Since $AD = AC$, we have $\angle ADX = \angle C$. Therefore, $\angle D < \angle ABC$. Since $\angle D = 90° - \angle DAX$ and $\angle ABC = 90° - \angle BAX$, we have $90° - \angle DAX < 90° - \angle BAX$, so $\angle BAX < \angle DAX$. Therefore, D is beyond B on $\overrightarrow{CB}$ as shown at right.



Since $\angle ABC$ is acute, $\angle ABD$ is obtuse. This tells us $AD > AB$. Since $AD = AC$, we have the desired $AC > AB$. □

Problem 10.5: Let $\triangle ABC$ be acute. Show that if $AC > AB$, then $\angle B > \angle C$.

Solution for Problem 10.5: What's wrong with this solution:

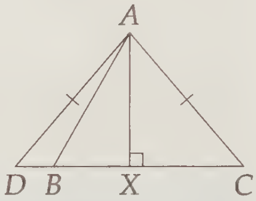
Bogus Solution: If $\angle B > \angle C$, then $AC > AB$. Since we know that $AC > AB$, we have $\angle B > \angle C$.

The problem with that 'proof' is that it is exactly the same as saying 'If I live in California, I live in America. Since I live in America, I live in California.' As we've noted before, we cannot assume that the converse of a true statement is true! We must prove a statement and its converse separately. Here, we will present two solutions.

Solution 1: Although our Bogus Solution was indeed quite bogus, we can still use what we already know about angles and sides. Specifically, consider three cases:

Case 1: $\angle B > \angle C$. As we have seen, if $\angle B > \angle C$, then $AC > AB$.
Case 2: $\angle B = \angle C$. If $\angle B = \angle C$, then $AC = AB$.
Case 3: $\angle B < \angle C$. As we have seen, if $\angle B < \angle C$, then $AC < AB$.

We are given that $AC > AB$. The only one of these cases that leads to $AC > AB$ is Case 1, so we can deduce that $\angle B > \angle C$. Make sure you see why this solution works, but the bogus one doesn't!



Solution 2: We can backtrack through our solution to Problem 10.4. We find point D on $\overrightarrow{CB}$ such that $AD = AC$. To show that D is beyond B on $\overrightarrow{CB}$, we first draw altitude $\overline{AX}$. Since we are given $AC > AB$ and we have both $AC^2 = AD^2 = AX^2 + XD^2$ and $AB^2 = AX^2 + XB^2$, we must have $XD > XB$.

Now we can use $\triangle ABD$ to show that $\angle C < \angle ABC$. From $\triangle ABD$, we have $\angle ADB + \angle ABD = 180^\circ - \angle DAB$. Therefore, $\angle ADB + \angle ABD < 180^\circ$. Since $\angle ADB = \angle C$ and $\angle ABD = 180^\circ - \angle ABC$, substitution into $\angle ADB + \angle ABD < 180^\circ$ gives us

$$\angle C + (180^\circ - \angle ABC) < 180^\circ,$$

so $\angle C < \angle ABC$ as desired. □

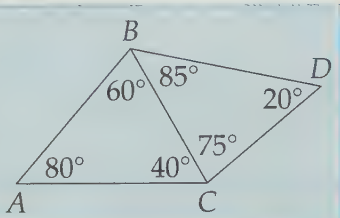
We can summarize all of our discoveries thus far in this section very simply:

Important: In any triangle, the longest side is opposite the largest angle and the shortest side is opposite the smallest angle. The middle side, of course, is therefore opposite the middle angle.

In other words, in $\triangle ABC$, $AB \geq AC \geq BC$ if and only if $\angle C \geq \angle B \geq \angle A$.

Let's try using these facts on a couple problems.

Problem 10.6: Given the angles as shown, order the lengths AB, BD, CD, BC, and AC from least to greatest. (Note: The diagram is not drawn to scale!)

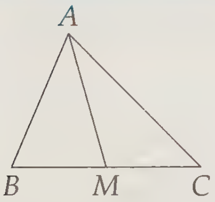


Solution for Problem 10.6: First we focus on $\triangle ABC$. Since $\angle A > \angle B > \angle C$, we have $BC > AC > AB$. Then, we turn to $\triangle BDC$. Since $\angle B > \angle C > \angle D$, we have $CD > BD > BC$. We got lucky! BC is the smallest in one inequality string and the largest in the other, so we can put the inequalities together:

$$CD > BD > BC > AC > AB.$$

□

Problem 10.7: In $\triangle ABC$ the median $\overline{AM}$ is longer than $BC/2$. Prove that $\angle BAC$ is acute.



Solution for Problem 10.7: We start with a diagram. We'd like to prove something about an angle, but all we are given is an inequality regarding lengths. So, we use the length inequality to get some angle inequalities to work with. Specifically, since $AM > BM$ in $\triangle ABM$ and $AM > MC$ in $\triangle ACM$ (because $AM > BC/2$ and $BM = MC = BC/2$), we have

$$\begin{aligned} \angle B &> \angle BAM \\ \angle C &> \angle CAM. \end{aligned}$$

We want to prove something about $\angle BAC$, which equals $\angle BAM + \angle CAM$, so we add these two inequalities to give $\angle B + \angle C > \angle BAC$. Since $\angle B + \angle C + \angle BAC = 180°$, we can write $\angle B + \angle C > \angle BAC$ as

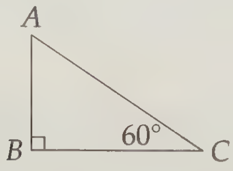
$$180° - \angle BAC > \angle BAC.$$

This gives us $180° > 2\angle BAC$, so $90° > \angle BAC$. Therefore, $\angle BAC$ is acute. □

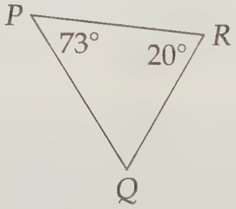
Exercises

10.1.1 In each of the parts below, order all the segments in the diagram from longest to shortest. (The diagrams are not drawn to scale!)

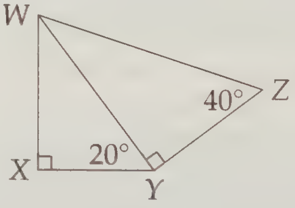
(a)



(b)



(c)



10.1.2 In isosceles triangle $\triangle PQR$, we have $\angle P = 54°$ and $PQ = PR$. Which is longer, $\overline{PQ}$ or $\overline{QR}$?

10.1.3★ Point X is on $\overleftrightarrow{AB}$. Point C is given such that $\angle AXC = \angle ACB = 100°$. Show that X is on segment $\overline{AB}$, rather than beyond B on $\overrightarrow{AB}$ or beyond A on $\overrightarrow{BA}$. **Hints:** 579, 483

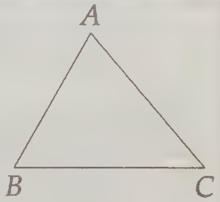
10.2 Pythagoras – Not Just For Right Triangles?

Back in Section 6.1, we learned that if $AC^2 + BC^2 = AB^2$, then $\angle ACB$ is a right angle. But what if $AC^2 + BC^2$ doesn't equal AB^2?

Problems

Problem 10.8: Draw lots of different triangles. Make a conjecture about what must be true if $AB^2+BC^2 > AC^2$, and what must be true if $AB^2 + BC^2 < AC^2$. Try to prove your conjectures before continuing!

Problem 10.9: Let $\angle C$ in $\triangle ABC$ be acute as shown. We will prove that



$$AC^2 + BC^2 > AB^2.$$

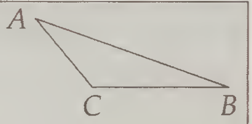
(a) Draw altitude $\overline{AX}$. Find an expression for AB^2.

(b) Show that $AC > AX$.

(c) Show that $AB^2 < AC^2 + BC^2$.

(d) Our diagram assumes that $\angle B$ is also acute. What if it isn't?

Problem 10.10: Let $\angle C$ in $\triangle ABC$ be obtuse as shown. We will prove that



$$AB^2 > AC^2 + BC^2.$$

(a) Draw altitude $\overline{AD}$ to $\overleftrightarrow{BC}$.

(b) Find an expression for AB.

(c) Find AD in terms of AC and CD, and find BD in terms of BC and CD.

(d) Use the previous two parts to show that $AB^2 > AC^2 + BC^2$.

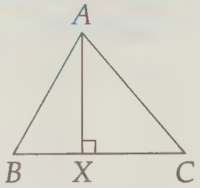
Problem 10.11: Prove that the converses of the facts we proved in Problems 10.9 and 10.10 are also true. In other words, prove that if $AB^2 > AC^2 + BC^2$, then $\angle C$ is obtuse, and prove that if $AB^2 < AC^2 + BC^2$, then $\angle C$ is acute.

Problem 10.12: In $\triangle XYZ$, we have $XY = 11$, $YZ = 14$. For how many integer values of XZ is $\triangle XYZ$ acute?

We know that if $\triangle ABC$ is a right triangle with hypotenuse $\overline{AB}$, we have $AC^2+BC^2 = AB^2$. Intuitively, it seems that if $\triangle ABC$ is acute, then $AC^2+BC^2 > AB^2$, and if it is obtuse with $\angle C > 90°$, then $AC^2+BC^2 < AB^2$. But intuitively isn't good enough for us, is it?

Problem 10.9: Let $\angle C$ in $\triangle ABC$ be acute. Prove that $AC^2 + BC^2 > AB^2$.

Solution for Problem 10.9: Those sums of squares of sides send us hunting for right triangles. Drawing altitude $\overline{AX}$ gives us a couple. From right triangle $\triangle ABX$, we have $AB^2 = AX^2 + BX^2$. Clearly $BX < BC$, and right triangle $\triangle ACX$ gives us $AX < AC$. Therefore, we have

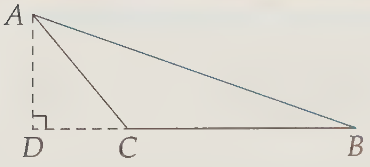


$$AB^2 = AX^2 + BX^2 < AC^2 + BC^2.$$

You might be wondering 'Where did we use the fact that $\angle C$ is acute?' Look closely – why must X be on $\overline{BC}$? If $\angle B$ and $\angle C$ are both acute, then altitude $\overline{AX}$ must meet $\overline{BC}$. Our proof, therefore, doesn't address the case where $\angle B$ is obtuse. However, in this case we have $AC > AB$, so obviously $AC^2 + BC^2 > AB^2$. □

Concept: One of the most useful approaches to solving new problems is to try to think of similar-looking problems you know how to handle. For example, our main step in the solution to Problem 10.9 was thinking to build right triangles once we saw the sum of squares of sides.

Problem 10.10: Let $\angle C$ in $\triangle ABC$ be obtuse. Prove that $AB^2 > AC^2 + BC^2$.



Solution for Problem 10.10: Once again, we start by drawing an altitude ($\overline{AD}$) to build right triangles. From right triangle $\triangle ABD$, we have $AB^2 = AD^2 + BD^2$. In order to get a BC term and an AC term in our equation, we note that $AD^2 = AC^2 - CD^2$ from $\triangle ADC$, and $BD = BC + CD$. Therefore, we have

$$\begin{aligned} AB^2 &= AD^2 + BD^2 \\ &= AC^2 - CD^2 + (BC + CD)^2 \\ &= AC^2 - CD^2 + BC^2 + 2(BC)(CD) + CD^2 \\ &= AC^2 + BC^2 + 2(BC)(CD) \end{aligned}$$

Since $2(BC)(CD)$ must be positive, we have $AB^2 > AC^2 + BC^2$. □

Make sure you see where in our solution we used the fact that $\angle ACB$ is obtuse.

Concept: If you don't use all the given information in a solution, proofread it closely to make sure you haven't made a mistake! Sometimes you won't need all the information you're given, but often if you haven't used it all, it's because you made a mistake somewhere.

Furthermore, when you're stuck on a problem, read it again and see if there is any information you haven't used yet!

In Problem 6.7 on page 138 we proved the converse of the Pythagorean Theorem; namely, that if the sides of a triangle satisfy $a^2 + b^2 = c^2$, then the triangle is right. Now we investigate if the converses of the inequalities we have just discovered are true.

Problem 10.11: Prove that the converses of the facts we proved in Problems 10.9 and 10.10 are also true. In other words, prove that if $AB^2 > AC^2 + BC^2$, then $\angle C$ is obtuse, and prove that if $AB^2 < AC^2 + BC^2$, then $\angle C$ is acute.

Solution for Problem 10.11: We can mirror our first solution to Problem 10.5 by considering three different cases:

Case 1: $\angle C$ is acute. If $\angle C$ is acute, then $AB^2 < AC^2 + BC^2$.
Case 2: $\angle C$ is right. If $\angle C$ is right, then $AB^2 = AC^2 + BC^2$.
Case 3: $\angle C$ is obtuse. If $\angle C$ is obtuse, then $AB^2 > AC^2 + BC^2$.

Each possible triangle $\triangle ABC$ must fall under exactly one of these cases. Only in the first case is $AB^2 < AC^2 + BC^2$, so if $AB^2 < AC^2 + BC^2$, we can conclude that $\angle C$ is acute. Similarly, only in the third case is $AB^2 > AC^2 + BC^2$, so we can conclude that $\angle C$ is obtuse whenever $AB^2 > AC^2 + BC^2$.

In the Exercises, you'll have a chance to provide a second 'geometric' argument, as we did in Solution 2 to Problem 10.5. □

We can now relate the squares of the sides of a triangle to the angles of the triangle:

Important: $\angle C$ of $\triangle ABC$ is acute if and only if $AB^2 < AC^2 + BC^2$.
$\angle C$ of $\triangle ABC$ is right if and only if $AB^2 = AC^2 + BC^2$.
$\angle C$ of $\triangle ABC$ is obtuse if and only if $AB^2 > AC^2 + BC^2$.

Let's try using some of this information on a problem.

Problem 10.12: In $\triangle XYZ$, we have $XY = 11$ and $YZ = 14$. For how many integer values of XZ is $\triangle XYZ$ acute?

Solution for Problem 10.12: In order for $\triangle XYZ$ to be acute, all three of its angles must be acute. Using the given side lengths and our inequalities above, we see that we must have

$$\begin{aligned} 121 + 196 &> XZ^2 \\ 196 + XZ^2 &> 121 \\ 121 + XZ^2 &> 196 \end{aligned}$$

Simplifying these three yields:

$$\begin{aligned} 317 &> XZ^2 \\ XZ^2 &> -75 \\ XZ^2 &> 75 \end{aligned}$$

The middle inequality is clearly always true. (We really didn't even have to include it – clearly 11 couldn't ever be the largest side!) Combining the other two and noting that we seek integer values of XZ, we have $9 \leq XZ \leq 17$ (since $8^2 < 75 < 9^2$ and $18^2 > 317 > 17^2$). So, there are 9 integer values of XZ such that $\triangle XYZ$ is acute. □

Exercises

10.2.1 Each of the following groups of three numbers are the lengths of the sides of a triangle. Identify each triangle as acute, right, or obtuse.

(a) 6, 8, 10

(b) 6, 8, 11

(c) 6, 8, 9

(d) $\sqrt{13}, 3\sqrt{3}, \sqrt{15}$

(e) 2.1, 1.8, 3.1

(f) 5/7, 9/14, 1

10.2.2 In $\triangle ABC$, $AB = 17$ and $BC = 27$. For how many integer values of AC is $\triangle ABC$ acute?

10.2.3★ In this problem we will find a geometric proof that $\angle ACB$ is obtuse if $AB^2 > AC^2 + BC^2$.

(a) Let X be the point on $\overleftrightarrow{BC}$ such that $\overline{AX}$ is an altitude of $\triangle ABC$. Find an expression for AB^2 in terms of other segment lengths in the resulting diagram. (Note that at this point, we do not know if X is on $\overline{BC}$ or not. It might be on the extension of the segment instead.)

(b) Why must $\triangle ABC$ be obtuse if X is not on $\overline{BC}$, but is instead on $\overleftrightarrow{BC}$ such that C is between B and X?

(c) If X is on $\overline{BC}$, then how are BX, CX, and BC related?

(d) Solve your expression from part (c) for BX and substitute the result into your equation for part (a). Is it possible for AB^2 to be greater than $AC^2 + BC^2$? **Hints:** 488

(e) Complete the proof by concluding that X must be on $\overleftrightarrow{BC}$ such that C is between X and B, so $\angle ACB$ must be obtuse. **Hints:** 169

10.3 The Triangle Inequality

The Triangle Inequality answers the question 'When can three given segment lengths be the side lengths of a triangle?' The answer to this question is both more obvious and more powerful than the other inequalities we have explored in this chapter. We'll start with a proof of this simple inequality, then show why the inequality is so powerful by using it to solve a variety of problems.

Problems

Problem 10.13: In this problem we will prove the Triangle Inequality. Don't skip over this problem because it looks obvious! Try to find a mathematical proof for each part.

(a) Use the inequalities from the previous sections to prove that in any triangle, the sum of two sides is greater than the third side. This is the Triangle Inequality.

(b) The Triangle Inequality is often written as 'the sum of two sides is greater than or equal to the third side.' Why – where does the 'or equal to' come from?

(c) If a, b, and c are positive numbers such that $a + b > c$, $a + c > b$, and $b + c > a$, then show that there exists a triangle with side lengths a, b, and c.

Problem 10.14: In how many ways can we choose three different numbers from the set $\{1, 2, 3, 4, 5, 6\}$ such that the three could be the sides of a triangle? (Note: The order of the chosen numbers doesn't matter; we consider $\{3, 4, 5\}$ to be the same as $\{4, 3, 5\}$.)

Problem 10.15: Can the lengths of the altitudes of a triangle be in the ratio $2 : 5 : 6$? Why or why not?

Problem 10.16: Let $\overline{AM}$ be a median of $\triangle ABC$. Prove that $AM > (AB + AC - BC)/2$.

Problem 10.17: Circle O and circle P are tangent at point T such that neither circle passes through the interior of the other. In this problem we will prove that O, P, and T are collinear (i.e., that a line passes through all three).

(a) Prove that $OT + TP \geq OP$.

(b) Show that if $OT + TP > OP$, then $\odot O$ and $\odot P$ meet at a second point besides T.

(c) Show that O, P, and T must be collinear.

We'll start by 'proving the obvious.'

Problem 10.13:

(a) Prove that in any triangle, the sum of two sides is greater than the third side. This is the **Triangle Inequality**.

(b) The Triangle Inequality is often written as 'the sum of two sides is greater than or equal to the third side.' Why – where does the 'or equal to' come from?

(c) If a, b, and c are positive numbers such that $a + b > c$, $a + c > b$, and $b + c > a$, then show that there exists a triangle with side lengths a, b, and c.

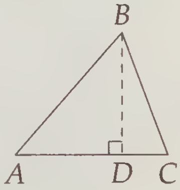
Solution for Problem 10.13:

(a) What's wrong with this solution:

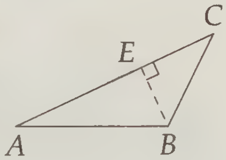
Bogus Solution: Suppose we draw $\overline{AB}$ first. Clearly the farthest C could be from A is if we just draw $\overline{BC}$ in the same direction, so that $AB + BC = AC$. Any other way we draw $\overline{BC}$ will result in C being closer to A. Therefore, $AB + BC > AC$ if $\triangle ABC$ is a real triangle.

The second sentence is true, but we haven't proved it. Since we're dealing with an inequality, we reach for the inequalities we've already proved to tackle the Triangle Inequality.

We've had so much success building right triangles, we do so again here, drawing altitude $\overline{BD}$ as shown. We have two cases to consider:



Case 1: $\triangle ABC$ is acute. We draw altitude $\overline{BD}$, forming two right triangles. We thus have $AB > AD$ and $BC > CD$. Adding these gives the desired $AB + BC > AD + DC = AC$. We can do the same for each angle, showing that each side is less than the sum of the other two sides.



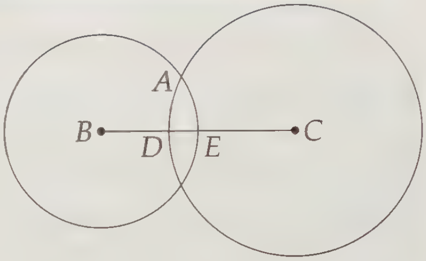
Case 2: $\triangle ABC$ is right or obtuse. If $\angle B \geq 90^\circ$ as shown, then clearly $AC > AB$ and $AC > BC$, so we definitely have $AB < AC + BC$ and $BC < AC + AB$. All we have left to prove is $AC < AB + BC$. Once again, right triangles come to the rescue. Drawing altitude $\overline{BE}$ gives us $AB > AE$ and $BC > EC$, so $AB + BC > AE + EC = AC$.

(b) We have seen above that if A, B, and C are not collinear, then there cannot be equality. However, when A, B, and C are collinear such that B is between A and C, we have $AB + BC = AC$. If A, B, and C are collinear, we call the resulting 'triangle' $\triangle ABC$ a **degenerate triangle**. (Typically the term 'triangle' only refers to **nondegenerate triangles**, i.e. those in which the vertices are not collinear.)

Note that when B is on $\overline{AC}$, we still have $AC + BC > AB$ and $AC + AB > BC$ in addition to $AB + BC = AC$, so the Triangle Inequality is still satisfied with a small modification:

> **Important:** The Triangle Inequality states that for any three points, A, B, and C, we have
> $$AB + BC \geq AC,$$
> where equality holds if and only if B is on $\overline{AC}$. Therefore, for nondegenerate triangles, $AB + BC > AC$.

(c) Suppose $a \geq b \geq c$, and let our triangle be $\triangle ABC$ with $AB = c$, $AC = b$, and $BC = a$. We can start building our triangle by drawing $\overline{BC}$ with length a. We know that A must be on the circle with center B and radius $AB = c$, and on the circle with center C and radius $AC = b$. We draw both circles. Because $a \geq b \geq c$, we know that $BC \geq AC \geq AB$. Therefore, the radius of $\odot B$ is less than BC, so $\odot B$ intersects $\overline{BC}$. We call this point of intersection E. Similarly, $\odot C$ meets $\overline{BC}$ at point D as shown.



Since $b + c > a$, we know that $BE + DC = b + c > a = BC$. Since $BE + DC > BC$, we know that D cannot be on $\overline{EC}$. In other words, our two circles must meet! Where these circles meet gives us our final vertex of $\triangle ABC$.

□

Note that we started off our proof to the last part with 'Suppose $a \geq b \geq c$.' But what if that's not the case? Does our whole proof fall apart?

No, it doesn't. The whole proof still works; we just have to rearrange the letters a little. All the same logic still works.

> **Important:** Mathematicians have a special way of saying 'All the different cases are essentially the same, so proving it for this one proves it for all of them.' To say this in our solution to the last part of the previous problem, a mathematician would have written, '**Without loss of generality**, let $a \geq b \geq c \ldots$.' Then, the mathematician doesn't need a whole separate proof for essentially equivalent cases such as $b \geq a \geq c$. Sometimes 'without loss of generality' is shortened to '**WLOG**'.

Now we'll use the Triangle Inequality to determine sets of specific side lengths that can be the sides of a triangle.

Problem 10.14: In how many ways can we choose three different numbers from the set $\{1, 2, 3, 4, 5, 6\}$ such that the three could be the sides of a nondegenerate triangle? (Note: The order of the chosen numbers doesn't matter; we consider $\{3, 4, 5\}$ to be the same as $\{4, 3, 5\}$.)

Solution for Problem 10.14: We first notice that if we have three numbers to consider as possible side lengths of a triangle, we only need to make sure that the sum of the smallest two is greater than the third. (Make sure you see why!) We could just start listing all the ones we see that work, but we should take an organized approach to make sure we don't miss any. We can do so by classifying sets of three numbers by the smallest number.

Case 1: Smallest side has length 1. No triangles can be made with three *different* lengths from our set if we include one of length 1.

Case 2: Smallest side has length 2. The other two sides must be 1 apart, giving the sets $\{2,3,4\}$, $\{2,4,5\}$, and $\{2,5,6\}$.

Case 3: Smallest side has length 3. There are only three possibilities and they all work: $\{3,4,5\}$, $\{3,4,6\}$, $\{3,5,6\}$.

Case 4: Smallest side has length 4. The only possibility is $\{4,5,6\}$, which works.

Adding them all up, we have 7 possibilities. □

We sometimes have to use some other tools in addition to the Triangle Inequality.

Problem 10.15: Can the lengths of the altitudes of a triangle be in the ratio 2 : 5 : 6? Why or why not?

Solution for Problem 10.15: We don't know anything about how the lengths of the altitudes of a triangle are related to each other. We do, however, know a whole lot about how the lengths of the sides of a triangle are related to each other. Therefore, we turn the problem from one involving altitudes into one involving side lengths. We let the area be K, let the side lengths be a, b, c, and the lengths of the altitudes to these sides be h_a, h_b, h_c, respectively. Therefore, we have $K = ah_a/2 = bh_b/2 = ch_c/2$, so the sides of the triangle have lengths

$$\frac{2K}{h_a}, \frac{2K}{h_b}, \frac{2K}{h_c}.$$

If our heights are in the ratio 2 : 5 : 6, then for some x, our heights are $2x$, $5x$, and $6x$. Then, our sides are

$$\frac{2K}{2x}, \frac{2K}{5x}, \frac{2K}{6x}.$$

However, the sum of the smallest two sides is then

$$\frac{2K}{5x} + \frac{2K}{6x} = \frac{12K}{30x} + \frac{10K}{30x} = \frac{22K}{30x},$$

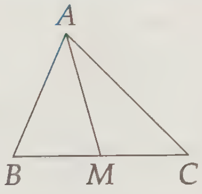
which is definitely less than the largest side, which is $2K/2x = K/x$. Therefore, the sides don't satisfy the Triangle Inequality, which means it is impossible to have a triangle with heights in the ratio 2 : 5 : 6.□

Concept: When facing problems involving lengths of altitudes of a triangle, consider using area as a tool.

The Triangle Inequality is probably the most commonly used tool in geometric inequality proofs; here's an example of the Triangle Inequality in action.

Problem 10.16: Let $\overline{AM}$ be a median of $\triangle ABC$. Prove that $AM > (AB + AC - BC)/2$.

Solution for Problem 10.16: We can apply the Triangle Inequality to both $\triangle ABM$ and $\triangle ACM$, giving:



$$\begin{aligned} AM + BM &> AB \\ AM + CM &> AC. \end{aligned}$$

Noting that $BM + CM = BC$, we think to add these inequalities, which gives $2AM + BC > AB + AC$. Subtracting BC from both sides and dividing by 2 yields

$$AM > \frac{AB + AC - BC}{2}.$$

□

Concept: If you don't see the path to the solution immediately, don't just sit and stare at the problem! Make some observations. Write down statements you can prove that might be helpful. Perhaps you'll be able to combine these observations to complete your proof.

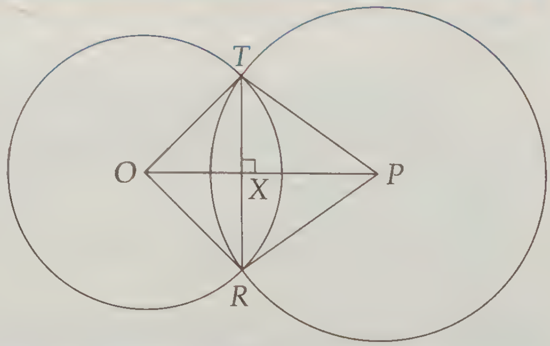
In Problem 10.16, even if we didn't see the solution immediately, the problem involves an inequality that has sums of lengths. This makes us think of using the Triangle Inequality to make observations. However, we don't simply write Triangle Inequality relationships blindly. We use the problem as a guide to make the observations. Specifically:

Concept: When trying to solve geometric inequalities, pay attention to which side of the inequality you want to prove each length is on. For example, we wouldn't want to start Problem 10.16 by observing $AM < AB + BM$, since this inequality has AM on the smaller side, but what we want to prove has AM on the larger side.

We end the chapter with an important application of the Triangle Inequality.

Problem 10.17: Circle O and circle P are tangent at point T such that neither circle passes through the interior of the other. Such circles are said to be **externally tangent**. Prove that O, P, and T are collinear.

Solution for Problem 10.17: We don't know a whole lot about tangent lines at this point. However, in this section we have discovered a way to show that three points are on a line. Specifically, if $OT + TP = OP$, then T is on $\overline{OP}$, because otherwise the Triangle Inequality guarantees $OT + TP > OP$.

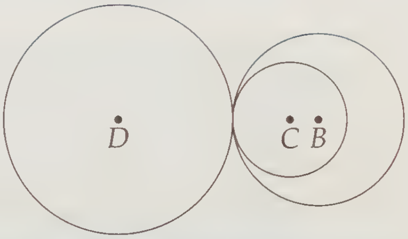


There doesn't seem to be any easy way to approach showing that $OT + TP = OP$. Instead, we try to show that it's impossible to have $OT + TP > OP$ and still have tangent circles. If we have $OT + TP > OP$, then T cannot be on $\overline{OP}$. Therefore, there is some point R such that $\overline{OP}$ is the perpendicular bisector of $\overline{TR}$. To find point R, we draw altitude $\overline{TX}$ from T to $\overline{OP}$, then we draw $\overline{RX}$ such that $RX = TX$ and $\overline{RX} \perp \overline{OP}$. Since $OX = OX$, $\angle TXO = \angle RXO$, and $RX = TX$, we have $\triangle TOX \cong \triangle ROX$. Therefore, $TO = RO$, so R is on $\odot O$. Similarly, we have $\triangle PTX \cong \triangle PRX$, so $PT = PR$ and R is on $\odot P$. Therefore, if $OT + TP > OP$, it is impossible for the circles to be tangent because there is a second point at which the circles meet.

However, our circles must be tangent, so we can't have $OT + TP > OP$. The Triangle Inequality tells us we can't have $OT + TP < OP$. Therefore, we must have $OT + TP = OP$, which means O, T, and P are collinear. □

Important: If $\odot O$ and $\odot P$ are tangent at point T, then O, P, and T are collinear. We'll often use this fact in problems with tangent circles by connecting the centers of the circles, knowing this line also goes through the point of tangency.

Notice that Problem 10.17 only tackles the case in which the circles are externally tangent. It's also true for circles that are **internally tangent**, i.e., when one circle is wholly inside the other except at the one point at which they are tangent. In the diagram at right, $\odot B$ and $\odot C$ are internally tangent, while $\odot D$ is externally tangent to both of the other circles.



Exercises

10.3.1 Which of the following sets of three numbers could be the side lengths of a triangle?

(a) $4, 5, 6$

(b) $7, 20, 9$

(c) $1/2, 1/6, 1/3$

(d) $3.4, 11.3, 9.8$

(e) $\sqrt{5}, \sqrt{14}, \sqrt{19}$

10.3.2 The lengths of two sides of a triangle are 7 cm and 3 cm. If the number of centimeters in the perimeter is a whole number, what is the number of centimeters in the positive difference between the greatest and least possible perimeters? *(Source: MATHCOUNTS)*

10.3.3 Prove that the sum of the diagonals of a quadrilateral is less than the quadrilateral's perimeter.

10.3.4★ $ABCD$ is a square and O is a point. Prove that the distance from O to A is not greater than the sum of the distances from O to the other three vertices, no matter which point we take to be point O. **Hints:** 97

10.3.5★ Prove that if a, b, and c are the sides of a triangle, then so are $\sqrt{a}$, $\sqrt{b}$, and $\sqrt{c}$. What about a^2, b^2, and c^2? **Hints:** 453, 36

Extra! *Calculating replaces thinking, while geometry stimulates it.*

–Jakob Steiner

10.4 Summary

Important: In any triangle, the longest side is opposite the largest angle and the shortest side is opposite the smallest angle. The middle side, of course, is therefore opposite the middle angle.

In other words, in $\triangle ABC$, $AB \geq AC \geq BC$ if and only if $\angle C \geq \angle B \geq \angle A$.

Important: $\angle C$ of $\triangle ABC$ is acute if and only if $AB^2 < AC^2 + BC^2$.
$\angle C$ of $\triangle ABC$ is right if and only if $AB^2 = AC^2 + BC^2$.
$\angle C$ of $\triangle ABC$ is obtuse if and only if $AB^2 > AC^2 + BC^2$.

Important: The Triangle Inequality states that for any three points, A, B, and C, we have

$$AB + BC \geq AC,$$

where equality holds if and only if B is on $\overline{AC}$. Therefore, for nondegenerate triangles (i.e., those in which the vertices are not collinear), $AB + BC > AC$.

Important: If $\odot O$ and $\odot P$ are tangent at point T, then O, P, and T are collinear. We'll often use this fact in problems with tangent circles by connecting the centers of the circles, knowing this line also goes through the point of tangency.

Problem Solving Strategies

Concepts:

- One of the most useful approaches to solving new problems is to try to think of similar-looking problems you know how to handle.
- If you don't use all the given information in a solution, proofread it closely to make sure you haven't made a mistake! Sometimes you won't need all the information you're given, but often if you haven't used it all, it's because you made a mistake somewhere.

  Furthermore, when you're stuck on a problem, read it again and see if there is any information you haven't used yet!
- When facing problems involving lengths of altitudes of a triangle, consider using area as a tool.

Continued on the next page. . .

Concepts: *. . . continued from the previous page*

- If you don't see the path to the solution immediately, don't just sit and stare at the problem! Make some observations – write down statements you can prove that might be helpful. Perhaps you'll be able to combine these observations to complete your proof.

- When trying to solve geometric inequalities, pay attention to which side of the inequality you want to prove each length is on. For example, we wouldn't want to start Problem 10.16 by observing $AM < AB + BM$, since this inequality has AM on the smaller side, but what we want to prove has AM on the larger side.

Review Problems

10.18 For each of the groups of three numbers below, state whether the numbers could be the side lengths of a triangle or not. If they can be, identify whether or not the triangle is acute, obtuse, or right.

(a) $2, 3, 4$

(b) $2.1, 1.7, 3.9$

(c) $\sqrt{5}, 2, \sqrt{3}$

(d) $199, 401, 297$ (See if you can do it without squaring those numbers!)

(e) $1/2, 1/3, 1$

(f) $60, 24, 48$

10.19 Ari is solving a problem involving a right triangle with legs 119 and 120. He uses the Pythagorean Theorem and gets 261 as the hypotenuse. He immediately shakes his head and starts over. How did he know so quickly that he made a mistake?

10.20 In $\triangle ABC$, $AB = 5$ and $BC = 11$. For which integer values of AC is $\triangle ABC$ an obtuse triangle?

10.21 $A_1A_2A_3\cdots A_n$ is a regular polygon with $n > 3$. Prove that $A_1A_3 > A_1A_2$.

10.22 Prove that it is impossible for the length of a side of a triangle to be greater than half the triangle's perimeter.

10.23 The perimeter of an isosceles triangle is 38 centimeters and two sides of the triangle are whole numbers in the ratio 3 : 8. What is the number of centimeters in the length of the shortest side? *(Source: MATHCOUNTS)*

10.24 Find all positive integers x for which it is possible for $2x + 3$, $3x + 8$, and $6x + 7$ to be the side lengths of a nondegenerate triangle.

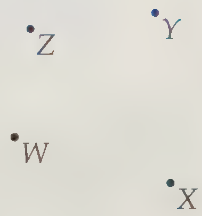
10.25 How many different obtuse triangles with integer side lengths and a perimeter of 20 can we draw such that no two of them are congruent?

10.26 The length of each leg of an isosceles triangle is $x+1$ and the length of the base is $3x-2$. Determine all possible values of x. (Your answer should be an inequality expressing the possible values of x.)

10.27 $\overline{YZ}$ is the base of isosceles triangle $\triangle XYZ$. Given that $YZ > XY$, show that $\angle X > 60^\circ$. **Hints:** 551, 209

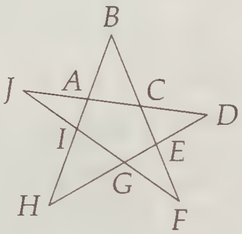
Challenge Problems

10.28 I have four stakes in my yard arranged like points W, X, Y, and Z shown. I wish to connect each stake to the same point in my yard with a string. What point should I choose to minimize the amount of string I must use? (Make sure to prove that your choice is the best possible!)

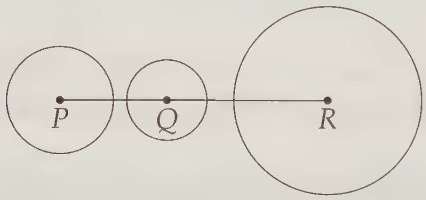


10.29 Orion chooses a positive number and Michelle chooses a positive number. Joshua chooses a positive number that is smaller than Orion's number. No matter what number Joshua chooses, we can always form a triangle with the three chosen numbers as side lengths. Show that Orion's and Michelle's numbers must be the same. **Hints:** 493

10.30 Show that it is impossible for any 5-pointed star like $ABCDEFGHIJ$ at right to have $AB > BC$, $CD > DE$, $EF > FG$, $GH > HI$, and $IJ > JA$ all be true.



10.31 As shown below, Q is on $\overline{PR}$ and circles $\odot P$, $\odot Q$, and $\odot R$ are drawn such that no two circles intersect and no one circle contains the other two. Suppose a fourth circle can be constructed that is externally tangent to all three of these circles.



(a) Use the Triangle Inequality to prove that the radius of our fourth circle is greater than the radius of $\odot Q$. **Hints:** 270

(b) Use the inequalities relating the order of sides in a triangle to the order of angles in a triangle to prove that the radius of our fourth circle is greater than the radius of $\odot Q$. **Hints:** 351

10.32 In convex quadrilateral $WXYZ$, we have $WX > WY$. Show that we must also have $XZ > YZ$. **Hints:** 447, 85

10.33 Let a, b, and c be three positive real numbers such that $a^2 + b^2 > c^2$, $a^2 + c^2 > b^2$ and $b^2 + c^2 > a^2$. Prove that a, b, and c can be the lengths of the sides of a triangle. **Hints:** 160

10.34 Prove that the sum of the lengths of the diagonals of a quadrilateral is greater than half the perimeter of the quadrilateral.

10.35★ In $\triangle XYZ$, we have $\angle X = 20^\circ$ and $XY = XZ$. Prove that $2YZ < XY < 3YZ$. **Hints:** 533, 84, 12, 395, 498

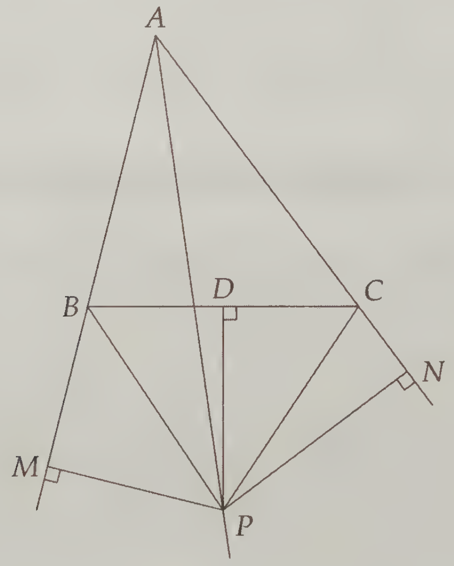
10.36★ In $\triangle ABC$ and $\triangle A'B'C'$, we have $AB = A'B'$, $AC = A'C'$, and $\angle BAC > \angle B'A'C'$. Prove that $BC > B'C'$. **Hints:** 266, 578

10.37★ Prove that the distance between any two points inside $\triangle ABC$ is not greater than half the perimeter of $\triangle ABC$. **Hints:** 430, 525, 437

Extra!

Proof That Every Triangle Is Isosceles?

In triangle ABC below, let D be the midpoint of $\overline{BC}$. Let P be the intersection of the angle bisector of A and the perpendicular bisector of $\overline{BC}$. Let M be the foot of the perpendicular from P to $\overline{AB}$ and N be the foot of the perpendicular from P to $\overline{AC}$, as shown.



Since P lies on the angle bisector of A, $\angle PAM = \angle PAN$. Also, $\angle PMA = \angle PNA = 90^\circ$, so by AAS, triangles PAM and PAN are congruent. Therefore, $PM = PN$ and $AM = AN$.

P lies on the perpendicular bisector of $\overline{BC}$, so $PB = PC$. Also, $\angle PMB = \angle PNC = 90^\circ$, so triangles PMB and PNC are congruent by HL Congruence. Therefore, $MB = NC$. But $AM = AB + BM$, $AN = AC + CN$, and $AM = AN$, so $AB = AC$. Therefore, triangle ABC is isosceles.

It is obviously not true that every triangle is isosceles, so something is going wrong in the proof. But what?